Manisha Choudhary, Ph.D.

Postdoctoral Research Associate
Department of Civil and Environmental Engineering
Cooperative Extension
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Education

Ph.D. Dec 2019	**Department of Chemical Engineering** **Indian Institute of Technology (IIT) Kharagpur, India** Dissertation: Water Treatment: Novel Materials and Methods for Sustainable Solution Advisor: Sudarsan Neogi
M.S. Jun 2014	**Department of Chemical Engineering** **Indian Institute of Technology (IIT) Kharagpur, India** Dissertation: Rice Mill Effluent Treatment: A Comparative Study And Extraction of Natural Coagulant Advisor: Sudarsan Neogi
B.S. Aug 2011	**Department of Chemical Engineering (Awarded Honors)** **MBM University, Jodhpur, India**

Appointments

Post-Doc
Dec. 2025 – Present

Civil and Environmental Engineering, and Cooperative Extension, University of Maine, Orono, ME

- Non-thermal plasma system for PFAS degradation from soil
- Regeneration of PFAS-laden granular activated carbon (GAC)
- Virtual Reality (VR) educational tool for understanding PFAS transport in soil
- PFAS+ Research Center development (Committee member for the laboratory design and procurement of equipment team)

Post-Doc
Oct. 2022 – Nov. 2025

Civil and Environmental Engineering, University of Maine, Orono, ME
Frontier Institute for Research in Sensor Technologies, University of Maine, Orono, ME

- Development of edge-functionalized graphene-based adsorbents for PFAS removal
- Understanding the effect of oxidation of graphene on its microwave reactivity for regeneration purpose
- Application of hierarchical structure-supported palladium nanocatalysts for PFAS degradation

- Development of a hydrogen sensor using hierarchical structure-supported palladium nanocatalysts
- Understanding the role of the size of superparamagnetic iron-oxide nanoparticles on its microwave reactivity for environmental applications
- Examining how the concentration of oxygen functional groups influences the thermal regeneration temperature of PFAS-laden GAC
- Teaching and outreach activities

Post-Doc Feb. 2021 – Oct. 2022	**Civil and Environmental Engineering, Temple University, Philadelphia, PA** - Evaluation and enhancement of existing analytical methods for active pharmaceutical ingredients using LC/MS/MS - Design and development of a continuous-mode pulsed electrical discharge plasma reactor - Development of a rotating spark gap switch to generate high-voltage pulses - USAID project on the creation of the Center of Excellence on water resiliency in Egyptian partner universities (Role: Project Manager) - Workshop organizer and instructor
Assistant Professor Sept 2019 – Dec 2020	**Department of Civil Engineering** **Dr. Radhakrishnan Institute of Technology (DRIT), Jaipur, India** - Teaching and mentoring undergraduate students - Departmental and Institutional services
Grad. Asst. 2014 – 2019 (During PhD)	**Department of Chemical Engineering,** **Indian Institute of Technology Kharagpur, India** - Synthesis of bio-coagulant using *Opuntia ficus-indica* (cactus) and utilized it for pre-treatment of oil sands process affected water from the tailing ponds of Syncrude Canada Ltd., Alberta, Canada - Synthesis of biochar and activated biochar for the removal of malachite green dye, Cu^{+2} and Ni^{+2} - Design and development of a multiple pin-plate pulsed electrical discharge plasma reactor for degradation of 17α-ethinylestradiol, Alachlor, Chlorpyrifos and four commercial reactive dyes - Photocatalytic degradation of phenol in water using ZnO nanopowder under UV-C light
Grad. Asst. 2012 – 2014 (During MS)	**Department of Chemical Engineering,** **Indian Institute of Technology Kharagpur, India** - Extraction of natural coagulant from Moringa oleifera and a comparative study of treatment of rice mill effluent by electrocoagulation and chemical coagulation
Student Intern May – Jun 2009	**Bhabha Atomic Research Center (BARC), Vashi, Navi Mumbai, India** - Unit operations in the opening of beryl ore for the beryllium metal extraction

Student Intern **Oil and Natural Gas Corporation Limited (ONGC), Nazira, Assam, India**
Jun – Jul 2008 - Operations during oil processing at the Geleki asset

Awarded and Submitted Research Grants

G9. **Choudhary, M.** (PI), **March 2026**, Graphene-Based Microwave-Regenerable 3D-Printed Adsorbents for PFAS Removal from Agricultural Waters, the U.S. Department of Agriculture, Agricultural Research Service, The University of Maine. ($66,000) (Accepted).

G8. Miller, R. (PI)**, Choudhary, M.** (Lead author and Co-PI), Adeleye, A. (Co-PI), Ravilla, A. (Co-PI), King, P. (Co-PI), Andrade, M. (Co-PI), **Feb 2026**, In Situ PFAS Sequestration and Removal from Maine Farm Soils Using Recoverable Adsorbents Compatible with Standard Farm Operations, Maine Department of Agriculture, Conservation and Forestry. ($499,834) (Submitted).

G7. **Choudhary, M.** (PI), Ravilla, A. (Lead author and Co-PI), Miller, R. (Co-PI), **Aug 2025**, Techno-Economic and Energy Analysis of Non-Thermal Plasma Regeneration of Granular Activated Carbon for PFAS Mitigation, the U.S. Department of Agriculture, Agricultural Research Service, The University of Maine. ($51,753) (Oct 2025 – July 2026)

G6. Zhao, X. (PI), **Choudhary, M.** (Lead author and Co-PI), **Aug 2025**, In-situ Regeneration of PFAS-Laden Granular Activated Carbon Using Non-Thermal Plasma: Optimizing Process Conditions and By-Product Characterization, the U.S. Department of Agriculture, Agricultural Research Service, The University of Maine. ($54,407) (Oct 2025 – July 2026)

G5. MacRae, J. (PI), **Choudhary, M.** (Lead author and Co-PI), Zhao, X. (Co-PI), Perry, R. (Co-PI), **2025**, A Virtual Reality Educational Tool for Understanding PFAS Transport in Soil, UMS TRANSFORMS, The University of Maine. ($75,000) (June 2025 – Dec 2027)

G4. Zhao, X. (PI), **Choudhary, M.** (Lead author and Co-PI), Apul, O.G. (Co-PI), **2025,** Energy-efficient PFAS Immobilization and Degradation in Soil using Non-thermal Plasma Electrodes coated with Activated Carbon, Maine Department of Agriculture, Conservation and Forestry ($285,000). (Aug 2025 – Sept 2027)

G3. Zhao, X. (PI), **Choudhary, M.** (Lead author and Co-PI), Apul, O.G. (Co-PI), MacRae, J. (Co-PI), **2024**, Lossless regeneration of PFAS-laden activated carbon by destruction of PFAS using non-thermal plasma technology at Maine's farms, the U.S. Department of Agriculture, Agricultural Research Service, the University of Maine. ($91,000) (April 2025 – April 2026)

G2. Apul, O.G. (PI), **Choudhary, M**. (Lead author and Co-PI), **2024**, Graphene-Enabled Water Treatment Technology for Targeted PFAS Removal from Natural Source Waters, the University of Maine's Innovation, Research, and Technology Accelerator (MIRTA 7.0) Program, The University of Maine. ($25,000) (Completed, Aug 2024 – Nov 2024)

G1. Suri, R.S. (PI), Biswas S.K. (Co-PI), Du L. (Co-PI), Tao R. (Co-PI), **Choudhary, M.** (Co-Author and Senior Personnel), **2021**, Catalytic Collaborative Funding Initiative – The CAT Program through the Office of the Vice President for Research (OVPR), Temple University, Philadelphia, for two years to research non-thermal plasma technology for water treatment (PFAS remediation). ($100,000) (Completed, April 2021 – Oct 2022)

Peer-reviewed Publications and Conference Presentations

J15. **Choudhary, M**., Martin, A., Alulema-Pullupax, P., Hatinoglu, D. M., Mensah, K., Moavenzadeh, S. G., Susan S., Apul, O.G., **March 2026**, Interdisciplinary Pedagogy for Environmental Chemistry Communication Through Art-Engineering Collaboration, *Journal of Chemical Education*. (Accepted)

J14. **Choudhary, M.**, Sarkar, P., Sharma, S. K., Zhao, X., Apul, O. G., Neogi, S., **Jan 2026**, Application of a multiple pin-plate non-thermal plasma reactor for degradation of alachlor and chlorpyrifos, *Environmental Science and Technology*. (Revision submitted)

J13. Alulema-Pullupax, P., Hatinoglu, D. M., Moavenzadeh, S. G., Mensah, K., **Choudhary, M.**, Ortiz, S., Apul, O. G., **Dec 2025**, Molecular connectivity indices and soil properties to predict sorption of per- and polyfluoroalkyl substances, *Environmental Science: Processes & Impacts*. (Accepted)

J12. Moavenzadeh, S. G., **Choudhary, M.**, Hannan, M., Hettiarachchi, G. M., Apul, O. G., **2025**, A critical review of per- and polyfluoroalkyl substances adsorption by soil, *Journal of Hazardous Materials: Organics*, 1, 1, 100001.

J11. Hatinoglu, D., Lee, S. S. S., **Choudhary, M.**, Lee, J., Attanayake, S. B., Hwang, K. Y., Detellem, D., Phan, M. H., Fortner, J., Apul, O. G., **2025**, Microwave heating of superparamagnetic iron-oxide nanoparticles toward environmental hyperthermia-based applications, *ACS Applied Materials & Interfaces*, 17, 35, 49775–49783.

J10. Hannan, M., Evrendilek, F., Leclair, D., **Choudhary, M.**, Mensah, K., Aeppli, C., Venkatesan, A., Apul, O. G., **2025**, Aftermath of a major firefighting foam spill in Brunswick, Maine: Spatiotemporal dynamics of per- and polyfluoroalkyl substances in the downstream aquatic environment, *Journal of Hazardous Materials Letters*, 6, 100150.

J9. Stewart, S., Yu, H., Andaluri, G., **Choudhary, M.**, **2025**, Bordoloi, A., Suri, R. P. S., Application of anthropogenic chemical and biological markers to characterize receiving urban waterways for untreated sanitary waste, *Journal of Water Process Engineering*, 71, 107207.

J8. **Choudhary, M**., Wang, W., Mensah, K., Mukhopadhyay, S. M., Apul, O. G., **2024**, Disruption of conjugated π-electron system of graphene oxides diminishes their microwave reactivity, *Langmuir*, 40, 51, 26824–26834. (Cover page article)

J7. White, L. R., Miner, J. N., McKinney, L. D., Pierce, L., Folley, A., Larrabee A., Scrapchansky L., Fessler, W., **Choudhary, M.**, Pouria, R., Zare, S., Perry E., Edalatpour, S., Apul, O. G., **2024**, Howell, C., Variable-area sensor permits near-continuous multi-point measurements of aqueous biological and chemical analytes, *Industrial & Engineering Chemistry Research,* 64, 1, 382–391.

J6. Apul, O. G., **Choudhary, M.**, **2024**, Interplay of adsorptive properties and dielectric reactivity of graphenes upon edge functionalization, *Chemical Engineering Science*, 296, 120274.

J5. **Choudhary, M.**, Sarkar, P., Sharma, S. K., Kajla, A., Neogi, S., **2021**, Quantification of reactive species generated in pulsed electrical discharge plasma reactor and its application for 17α-ethinylestradiol degradation, *Separation and Purification Technology*, 275, 119173.

J4. **Choudhary, M.**, Kumar, R., Neogi, S., **2020**, Activated biochar derived from *Opuntia ficus-indica* for the efficient adsorption of malachite green dye, Cu^{+2} and Ni^{+2} from water, *Journal of Hazardous Materials*, 392, 122441.

J3. **Choudhary, M.**, Ray, M. B., Neogi, S., **2019**, Evaluation of the potential application of cactus (*Opuntia ficus-indica*) as a bio-coagulant for pre-treatment of oil sands process-affected water, *Separation and Purification Technology*, 209, 714–724.

J2. **Choudhary, M.**, Neogi, S., **2017**, A natural coagulant protein from *Moringa Oleifera*: Isolation, characterization, and potential use for water treatment, *Materials Research Express*, 4 105502.

J1. **Choudhary, M.**, Majumder, S., Neogi, S., **2015**, Studies on the treatment of rice mill effluent by electrocoagulation, *Separation Science and Technology*, 50, 505–511.

Under preparation

J16. **Choudhary, M.,** Neogi, S., Application of a multiple pin-plate non-thermal plasma reactor for degradation of commercial reactive dyes. (Ready for submission to *Environmental Science & Technology* Journal)

Oral and Poster Presentations

C18. **Choudhary, M**., Wang, W., Mensah, K., Mukhopadhyay, S.M., Apul, O. G., Oxidation of graphene disrupts its conjugated π-electron network resulting in reduced microwave reactivity. Environmental Nanotechnology Gordon Research Conference, USA, June 1 - 6, **2025**. (Poster presentation)

C17. **Choudhary, M**., Wang, W., Mensah, K., Mukhopadhyay, S.M., Apul, O. G., Oxidation of graphene disrupts its conjugated π-electron network resulting in reduced microwave reactivity. Environmental Nanotechnology Gordon Research Seminar, USA, May 31 - June 1, **2025**. (Oral Presentation)

C16. Hatinoglu, D., **Choudhary, M.**, Turzo, P., Loganathan, S., Hanigan, D., Apul, O. G., Understanding the effect of oxygen content on thermal regeneration of granular activated carbon as a step toward PFAS destruction. AEESP Research and Education Conference, Durham, NC. May **2025** (Poster Presentation)

C14. Hatinoglu, D., Lee, S.S., **Choudhary, M**., Lee J., Attanayake, S.B., Hwang, K.Y., Evrendilek, F., Phan, M.H., Fortner, J., Apul, O. G., Particle size-dependent magnetic and microwave hyperreactivity in superparamagnetic iron-oxide nanoparticles. NSF Nanoscale Science and Engineering Grantees Conference, Alexandria, VA. December **2024**. (Poster Presentation) (Best poster award)

C13. McCarthy, M., Mensah, K., **Choudhary, M.**, Moavenzadeh, S., Apul, O. G., Computational and theoretical modelling of nanobubble-graphene interactions towards understanding phenanthrene adsorption mechanism. North Carolina State University Future Leaders in Chemical Engineering Symposium **2024**. (Oral and Poster presentation)

C12. McCarthy, M., Mensah, K., **Choudhary, M.**, Moavenzadeh, S., Apul, O. G., Nanobubble and graphene interactions during phenanthrene adsorption: towards removal of dimethylsilanediol from wastewater on the International Space Station. Eckhardt Northeast Student Regional AIChE Conference **2024**. (Poster presentation) (Second prize winner).

C11. McCarthy, M., Mensah, K., **Choudhary, M**., Moavenzadeh, S., Apul, O. G., The Role of nanobubbles in phenanthrene adsorption: towards removal of dimethylsilanediol from wastewater on the International Space Station. the University of Maine Student Symposium, April **2024**. (Poster Presentation)

C10. Moavenzadeh Ghaznavi, S., **Choudhary, M.**, Apul, O. G., Partitioning of per- and polyfluoroalkyl substances in soil horizons impacted by biosolids. American Chemical Society Conference (ACS Fall 2024), Denver, CO, United States, **2024**. (Oral Presentation)

C9. Moavenzadeh Ghaznavi, S., **Choudhary, M.**, Apul, O. G., Investigation of per- and polyfluoroalkyl substances (pfas) adsorption mechanisms onto soil with a history of sludge land application in Maine. 2024 AEHS Foundation Conference, San Diego, California, March **2024**. (Poster presentation)

C8. **Choudhary, M.**, Wang, W., Mukhopadhyay, S.M., Apul, O. G., Synthesis of edge-tailored graphene oxide for removal of organic contaminants from water and its regeneration through microwave heating. Responding Together to Global Challenges, AEESP Research and Education Conference, Boston, MA, June 19-23, **2023.** (Oral Presentation)

C7. **Choudhary, M.**, Wang, W., Mukhopadhyay, S.M., Apul, O. G., Atomically precise tailoring of graphene nanosheets to control properties that are expedient for water treatment. Environmental Nanotechnology Gordon Research Conference, USA, June 4 - 9, **2023**. (Poster Presentation)

C6. Hafeez, B., Jennings, E., **Choudhary, M.**, Apul, O. G., Marangoni, T., Battigelli, A., Functionalized graphene oxide for water treatment. the University of Maine Student Symposium, April **2023**. (Poster Presentation)

C5. Moavenzadeh, S.G., **Choudhary, M.**, Apul, O. G., Kopec, D., Evaluating available options for sustainable management of PFAS in wastewater sludge in Maine: Searching for solutions to a wicked problem. Maine Sustainability and Water Conference, March **2023**. (Poster Presentation)

C4. **Choudhary, M.**, Neogi, S., **2017**, Evaluation of the efficacy and mechanism of cactus (Opuntia ficus-indica) as a natural coagulant for pre-treatment of oil sands process affected water. AIChE Annual Meeting 2017, Minneapolis, Minnesota, USA, October 29 - November 3, **2017**.

C3. **Choudhary, M.**, Neogi, S., **2016**, Photocatalytic degradation of phenol in water using ZnO nanopowder under UV-C light. International Conference on Water: from Pollution to Purification, ICW 2016, Kottayam, Kerala, India, December 12 - 15, **2016**. (Second prize winner)

C2. **Choudhary, M.**, Majumder, S., Neogi, S., Treatment of rice mill effluent by electrocoagulation. International Conference on Advances in Chemical Engineering, ICACE 2013, NIT Raipur, India, **2013**.

C1. **Choudhary, M.**, Majumder, S., Neogi, S., The role of electrode material in rice mill effluent treatment by electrocoagulation. Chemference 2012, ICT Mumbai and IIT Bombay, India, December 10 – 11, **2012**.

Invited Talks

I4. **Choudhary, M.**, Bridging Disciplines for a Cleaner and Sustainable Future: A Journey from Chemical Engineering to Environmental Solutions. Towson University, Towson, MD, USA, Oct **2025**.

I3. **Choudhary, M.**, Wang, W., Mensah, K., Mukhopadhyay, S.M., Apul, O.G., Role of Oxidation in Modifying Graphene's π-Electron Conjugation and Its Impact on Microwave Reactivity and Adsorption Capacity. The Geological Society of America Connects 2025, TX, USA, Oct 19 – 22, **2025**.

I2. **Choudhary, M.**, Wang, W., Mensah, K., Mukhopadhyay, S.M., Apul, O.G., Oxidation of Graphene Disrupts Its Conjugated π-Electron Network Resulting in Reduced Microwave Reactivity. Environmental Nanotechnology, Gordon Research Seminar, ME, USA, May 31 – Jun 1, **2025**.

I1. **Choudhary, M.**, Advancing Water Treatment and Science Communication to Mitigate Environmental Contamination in the Anthropocene Epoch. Union College, Schenectady, NY, USA, Feb **2025**.

Patents

P1. Apul, O.G.**, Choudhary, M**., Mensah, K., **2024**. Binder-free palletization of graphene nanosheets towards PFAS removal from drinking water sources. (Disclosed on Oct 07th, 2024)

Teaching and Mentorship Experience

1. **University of Maine, Orono, ME**
   - Instructor of Undergraduate Level Course, CIE 434 – Wastewater Process Design (with lab), Spring 2024, 2025
   - Guest Lecture, Graduate Level Course, CIE 598 – Environmental Nanotechnology, Spring 2025
   - Guest Lecture, Undergraduate Level Course, CIE 430 – Water Treatment, Fall 2024
   - Mentoring Experience
     - Patatri Chakraborty, PhD student
     - Sonia Moavenzadeh, PhD student
     - Dilara Hatinoglu, PhD student
     - Sathvik Peddamalla, PhD student
     - Sanskar Shrestha, MS student

  - Hayden Sawyer, UG
  - Samuel Robinson, UG student
  - Madi McCarthy, UG student
  - Aadarsh Lakshmanan, High school student
- Coordinator – Graduate Student Seminar Series

2. Temple University, Philadelphia, PA

- Mentoring Experience: Azita Hassan Mazandarani, PhD student
- Instructor and Coordinator: Water Quality Testing Workshop – Conventional Parameters, Aug. 2022
- Instructor and Coordinator: Water Quality Testing Workshop – Advanced Analytical Analysis, July 2022
- Instructor and Coordinator: Water Quality Testing Workshop, Dec. 2021

3. IIT Kharagpur, Kharagpur, India

- Mentoring Experience
  - Kanchan Nath, MS student
  - Puja Paul, MS student
  - Shubhi Paliwal, MS student
  - Sambhojit Roy, MS student
  - Rahul Kumar, UG student
  - Aditya Singh, UG student
  - Manoj Shinde, UG student
  - Shubham Kumar, UG student
  - Dibya Prakash Sethi, UG student
- Teaching Assistant: Graduate Student Seminar 2018, 2019
- Teaching Assistant: Project Engineering & Management, 2015, 2017
- Teaching Assistant: Industrial Pollution Control, 2014, 2016
- Coordinator and Instructor: Workshop on Mendeley and Endnote, Sep. 2016
- Coordinator: Research Scholar Day (2014 – 2018)
- Course Coordinator: Nanobiotechnology: A discipline at a crossroads
  (International Summer and Winter Term Course, Dec. 2015)
- Course Coordinator: Advanced Technologies for Wastewater Treatment and Recycling
  (International Summer and Winter Term Course, July 2014)
- Course Coordinator: Advanced Plasma Processing: Fundamentals and Applications
  (International Summer and Winter Term Course, June 2014)

Awards and Honors

1. Trailblazers in Engineering (TBE) Fellow, College of Engineering, Purdue University, **2025**
   (Selected as one of ~35 nationwide candidates for a competitive program designed to prepare outstanding future engineering faculty) (Link)

2. 40 Under 40 The Rising Stars in Environmental Engineering and Science American Academy of Environmental Engineers and Scientists (AAEES), **2024** (Link)
3. Awarded an international conference travel grant (equivalent to around $3000) at IIT Kharagpur, **2017**
4. Winner of second prize for the poster presentation at International Conference on Water: From Pollution to Purification, ICW 2016, Kottayam, Kerala, India, **2016** (during PhD)
5. Received Scholarship/Fellowship from the Ministry of Human Resource Development (MHRD), India, **2014-2019** (during PhD)
6. Achieved a silver medal for solving a problem statement provided by Oil and Natural Gas Corporation (ONGC), India, in the interahall General Championship event "Cheminnovation-2014" organized by IIT Kharagpur, India, **2014** (during masters)
7. Qualified Graduate Aptitude Test for Engineering (GATE) **2012** (Top 0.5%)
8. Achieved second prize for providing innovation in the utility of solar energy under the event "Solstice" in a national-level technical festival named "Neuron-2008" held at Malaviya National Institute of Technology (MNIT), Jaipur, India, **2008** (during undergraduate)
9. Awarded a scholarship named "Gargi Puraskar" from the Board of Secondary Education, Rajasthan, India, for academic merit in high school, **2004**

Professional Service and Involvement

- **Reviewer:** Springer Nature (Earth and Environmental Sciences Community), Journal of Hazardous Materials, Journal of Water Process Engineering, RSC Advances, Bioresource Technology, Environmental Chemical Engineering, Chemical Engineering Science, Environmental Research Letters, Separation Science and Technology, Biomass Conversion and Biorefinery, Chemosphere
- **National Science Foundation (NSF, USA) ad hoc Proposal Reviewer:** ECosystem for Leading Innovation in Plasma Science and Engineering (ECLIPSE) program (**2025**)
- **Proposal Reviewer:** Illinois-Indiana Sea Grant, Purdue University (**2025**)
- **Professional Memberships:** Association of Environmental Engineering and Science Professors (AEESP), Water Environment Federation (WEF), Indian Institute of Chemical Engineers (IIChE)
- **Established Program to Stimulate Competitive Research (EPSCoR) Representative from UMaine at NSF, USA: Engineering EPSCoR Engagement Convocation** (**2024**)
- **Assisted** in the preparation of **the ABET Self-Study Report** for the Department of Civil and Environmental Engineering, The University of Maine (**2024**)
- **Session Organizer and Instructor:** Sustainable Engineering Leaders of the Future (SELF) summer camp for high school girls (**2024**)
- **Workshop Instructor:** "Introduction to Water Pollution and the Importance of Water and Wastewater Treatment," for The Northern Region Girl Scouts (**2023**)

- **Mentored** five students from Greely High School, Cumberland, ME, in developing a filtration system for PFAS as part of Samsung's Solve for Tomorrow science competition; the team won the state championship for their innovative solution. (**2023**)
- **Session Co-Chair:** "PFAS: Characterization," Association of Environmental Engineering and Science Professors (AEESP), Research and Education Conference (**2023**)
- **Poster Judge Committee member:** Association of Environmental Engineering and Science Professors (AEESP), Research and Education Conference **(2023**)
- **Water For People (NGO):** Technical volunteer (**2022 – present**)
- **Gopali Youth Welfare Society (NGO to serve underserved communities):** Child Education Committee member (**2014 – 2019**)
- **Secretary General:** Managed the hall library in Sister Nivedita Hall of Residence at IIT Kharagpur (**2017**)
- **Selected Media Coverage:**
  - **UMaine: L.L.Bean supporting UMaine's work on eradicating 'forever chemicals'**
  - **Portland Press Herald: Greely Team State Champs with PFAS Project**
  - **WABI: UMaine brings art and science together to share an important message**
  - **News Center Maine: UMaine students showcase research on microplastics in the environment with art**